Incorporating Statistical Machine Translation Word Knowledge Into Neural Machine Translation

Xing Wang, Zhaopeng Tu, and Min Zhang

***Abstract*—Neural machine translation (NMT) has gained more and more attention in recent years, mainly due to its simplicity yet state-of-the-art performance. However, previous research has shown that NMT suffers from several limitations: source coverage guidance, translation of rare words, and the limited vocabulary, while statistical machine translation (SMT) has complementary properties that correspond well to these limitations. It is straightforward to improve the translation performance by combining the advantages of two kinds of models. This paper proposes a general framework for incorporating the SMT word knowledge into NMT to alleviate above word-level limitations. In our framework, the NMT decoder makes more accurate word prediction by referring to the SMT word recommendations in both training and testing phases. Specifically, the SMT model offers informative word recommendations based on the NMT decoding information. Then, we use the SMT word predictions as prior knowledge to adjust the NMT word generation probability, which unitizes a neural network based classifier to digest the discrete word knowledge. In this paper, we use two model variants to implement the framework, one with a gating mechanism and the other with a direct competition mechanism. Experimental results on Chinese-to-English and English-to-German translation tasks show that the proposed framework can take advantage of the SMT word knowledge and consistently achieve significant improvements over NMT and SMT baseline systems.**

***Index Terms*—Neural machine translation, statistical machine translation, hybrid translation, translation combination.**

I. Introduction

DEEP neural networks have been successfully applied in various research areas, such as image processing [1], speech recognition [2]–[4] and information retrieval [5]. In the field of machine translation, neural machine translation has recently achieved much progress in both academic and industrial fields [6]–[11]. It improves performance of both spoken language translation [12] and written language translation. NMT models the translation process by using an encoder-decoder architecture, which is widely used in sequence transformation tasks, e.g., image caption [13], text summarization [14], natural language conversation [15]. Specifically, the encoder-decoder architecture consists of two neural networks: a neural network-based encoder encodes a source sentence into a vector representation, based on which another neural network-based decoder decodes the vector representation to generate a variable-length target sentence word by word. NMT has several advantages over the conventional SMT models [16]–[21], which exploit elaborate features under the log-linear framework. First, the NMT model, which is essentially a conceptually single large neural network, is completely constructed by learning the translation knowledge from the training corpus with less effort of feature designed by linguists or engineers. Second, gating [22] and attention [9] techniques, which are widely adopted in the field of Natural Language Processing (NLP), have been proved to be effective in capturing potential long-distance dependencies and complicated word alignment information in the machine translation process, both of which pose serious challenges for SMT. Lastly, the memory consumption of the NMT model is much smaller than that of conventional SMT models, which maintain a large translation model, a reordering model and a language model [23].

Despite these advantages, recent studies find that NMT models generally generate fluent but sometimes inaccurate translation sequences, due to the following limitations:

1) *Coverage problem* [24]: NMT finishes target sentence generation when producing an end-of-sentence symbol 'EOS'. In the sentence generation process, NMT has the coverage problem which cannot guarantee that all source words are properly tarnslated, resulting in either "over-translation" or "under-translation" [24].
2) *Imprecise translation problem* [25]–[27]: The NMT decoder is essentially a language model conditioned on the source sentence representation. Accordingly, NMT generally produces words which are deemed to be very fluent but they may not convey the original meanings of the encoded source sentence.
3) *UNK problem* [23], [28]: Due to the expensive computational cost of target word prediction in the softmax layer, NMT uses a limited vocabulary to record the most frequent words and uses a UNK symbol to represent other words in both language sides. The occurrences of UNK symbols pose difficulties in translating these low-frequency words.

Compared to NMT, SMT models adopt different translation mechanism that can properly alleviate the above drawbacks:

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TABLE I
TRANSLATION EXAMPLES GENERATED BY SMT AND NMT SYSTEMS

SRC	去年 , 珠 三角 高 新 技术 产业带 的 产品 出口 合计 达 219.9亿 美元 , 占 全 省 高 新 技术 产品 出口额 的 98.7% 。
SMT	last year , the exports of high and new technology industrial belts in the prd reached 219.9亿 us dollars , were 98.7% percent of the province 's exports of high and new technology products.
NMT	last year , the export of high - tech and high - tech industrial belts in the prd region reached UNK billion us dollars , accounting for UNK percent of the total value of new high - tech products.
SRC	今年 前 两 月 广东 高新 技术 产品 出口 37.6亿 美元
SMT	37.6亿 us dollars during the first two months of this year guangdong high technology exports
NMT	hong kong exports UNK billion dollars in the first two months of this year.
SRC	拉斯穆森 说 : 我们 同样 认为 通过 对话 宽容 来 解决 事端 , 而 不 是 暴力 。
SMT	rasmussen said : we also believe resolved through dialogue tolerance , instead of violent incidents.
NMT	UNK said : " we also believe that we can resolve the incident through dialogue , but not violence.

1) The SMT decoder finishes the translation process when all source segments (words or phrases) are covered by the translation rules. It guarantees that each word in source sentence has been translated.
2) Words are treated as discrete symbols rather than continuous real-valued vectors in SMT, which ensures SMT models translate a source word to a semantic-related target word, alleviating imprecise translation problem.
3) Words are explicitly memorized in SMT translation rules. Therefore there are no UNK symbols in the SMT translation rules.

Table I shows translation examples generated by the SMT and NMT systems. The first example shows the over-translation problem (coverage problem), in which the Chinese phrase "Insert Graphics" appears two times in source sentence but the corresponding target phrase "high - tech" appears three times in the NMT output. The second example shows the imprecise translation problem, where the Chinese phrase "Insert Graphics" in the source sentence is mistakenly translated to "hong kong" in the NMT translation. The last one shows the UNK problem, in which the person name "Insert Graphics" is mapped into "UNK" due to the limited vocabulary size in NMT.

It is straightforward to leverage the full advantages of the two kinds of models for improving translation performances. In order to achieve this goal, several techniques have been proposed to augment NMT mdoels by utilizing SMT features or SMT outputs. For example, to alleviate the NMT limitations, [29] used the log-linear framework as a backbone and combined SMT features with the NMT model in the beam search. [30] proposed to utilize structured search spaces produced by a hierarchical phrase-based SMT model (Hiero) to guide NMT decoding by expanding the NMT search space with Hiero translation hypotheses.

This work follows the above research line and moves a step forward along the same direction. Specifically, we propose a general framework that incorporates the SMT word knowledge into the NMT encoder-decoder architecture. Specifically, first we train the SMT model on a parallel data by using the commonly used SMT approaches [17]. Then we use the SMT word knowledge as prior knowledge to adjust the NMT word generation probability, by utilizing a neural network-based classifier to digest the discrete word knowledge. The proposed neural network-based model makes the NMT model automatically learn to make use of the SMT knowledge, without having to heuristically combine NMT model with SMT features or SMT outputs.

A major challenge here is that the NMT and the SMT models generate the target translation sequence in different manners at decoding: the NMT decoder uses continuous distributed representation to generate the target sequence word by word, while the state-of-the-art SMT decoder utilizes the handcrafted features to generate target sequence phrase by phrase. Different representation approaches and different decoding granularities make the two kinds of models incompatible with each other. In this work, we make several modifications on the SMT decoder (see Section III) and only use the SMT word knowledge, to make the SMT model compatible with the NMT model.

To incorporate the SMT word knowledge into NMT neural network-based architecture, an additional neural network-based classifier is applied to score the word recommendations from the SMT model and estimate their probabilities, then these word recommendation probabilities are utilized to adjust the NMT word prediction probabilities. In this work, we use two model variants: 1) a gating mechanism and 2) a direct competition mechanism (softmax function) to estimate the probabilities of SMT word recommendations and NMT word prediction. Finally, we assign the adjusted word probabilities to the decoder for the next word generation. In addition, to address the UNK issue as described above, we propose to choose a proper word from SMT word recommendations by jointly considering the NMT attention history and the SMT coverage mechanism, and copy it to replace the UNK symbol in the generated sequence.

Experimental results on Chinese-to-English and English-to-German translation tasks show that the proposed framework, which incorporates the SMT word knowledge into the NMT encoder-decoder architecture, achieves significant improvements over the baseline NMT (RNNSEARCH) and SMT (MOSES) systems.

This article is a substantial extension of our conference paper [31] with the following new contents: (1) a new model architecture, (2) a comprehensive description of our models, (3) empirical verification of the proposed models on large scale Chinese-to-English and English-to-German translation tasks.

Roadmap The rest of this article is organized as follows. Section II briefly introduces the attention-based NMT and phrase-based SMT as the background. Section III presents our proposed model which incorporates the SMT model into NMT with attention mechanism. Section IV details the training processes of the proposed model. Section V describes a series of experiments that verify the effectiveness of our approach

on Chinese-to-English and English-to-German translation tasks and reports the experimental results. Finally we discuss the related work and conclude the article in Section VI and Section VII respectively.

II. Background

In this section, we present some background knowledge about NMT and SMT. Specifically, in the following subsections, we give a brief introduction to a typical NMT model, namely the attention-based NMT model, and the most popular SMT model, namely the phrase-based SMT model.

A. Attention-Based Neural Machine Translation

The attention-based NMT [9] adopts the encoder-decoder architecture: a neural network-based encoder encodes the whole source sentence $\mathbf{x} = x_1, x_2, \ldots, x_{T_x}$ into a sequence of vectors, then a neural network-based decoder uses these vectors to generate the target translation $\mathbf{y} = y_1, y_2, \ldots, y_{T_y}$.

a) Encoder: The encoder is generally implemented as a bidirectional Recurrent Neural Network (RNN), which uses a forward and a reverse RNNs [32] to encode the source sentence: the forward one takes the source sentence $\mathbf{x}$ as the input from left to right, computing a forward hidden states $\overrightarrow{h} = [\overrightarrow{h_1}, \overrightarrow{h_2}, \ldots, \overrightarrow{h_{T_x}}]$, the reverse one takes the source sentence $\mathbf{x}$ as input from right to left, calculating backward hidden states $\overleftarrow{h} = [\overleftarrow{h_1}, \overleftarrow{h_2}, \ldots, \overleftarrow{h_{T_x}}]$. Then forward hidden states and backward hidden states are concatenated to form annotations of the source sentence $\mathbf{h} = [h_1, h_2, \ldots, h_{T_x}]$, where

$$h_j = \left[\overrightarrow{h_j}; \overleftarrow{h_j}\right] \tag{1}$$

b) Decoder: The decoder essentially is a conditional recurrent language model which models the target sentence by decomposing into the generation of each target word:

$$\log p\left(\mathbf{y}|\mathbf{x}\right) = \sum_{t=1}^{T_y} \log p\left(y_t|\mathbf{y_{<t}}, \mathbf{x}\right) \tag{2}$$

Specifically, the decoder generates the word y_t after generating the partial sequence $\mathbf{y_{<y}} = y_1, y_2, \ldots, y_{t-1}$. The probability of y_t is calculated using the current decoding hidden state s_t, the previous generated target word y_{t-1} and the current context vector c_t:

$$p(y_t|\mathbf{y_{<t}}, \mathbf{x}) = softmax(f(s_t, y_{t-1}, c_t)) \tag{3}$$

where $f(\cdot)$ is a non-linear activation function and the current decoding hidden state s_t is computed on the previous decoding hidden state s_{t-1}, the previous generated target word y_{t-1} and the current context vector c_t as

$$s_t = g(s_{t-1}, y_{t-1}, c_t) \tag{4}$$

where $g(\cdot)$ is a a recurrent activation function, such as Long Short-Term Memory (LSTM) [22] or Gated Recurrent Unit (GRU) [7]. c_t is a time-dependent context representation, which is calculated as a weighted sum of source annotations:

$$c_t = \sum_{j=1}^{T_x} \alpha_{t,j} h_j \tag{5}$$

where the weight $\alpha_{t,j}$ of the source annotation h_j is calculated as

$$\alpha_{t,j} = \frac{\exp(e_{t,j})}{\sum_{k=1}^{T_x} \exp(e_{t,k})} \tag{6}$$

and

$$e_{t,j} = a(s_{t-1}, h_j) \tag{7}$$

is an attention model which measures how well the target word y_t and the source word annotation h_j match each other.

B. Statistical Machine Translation

Most SMT models are defined within the log-linear framework [33]. Given a source sentence $\mathbf{x}$, SMT searches for the best translation $\hat{\mathbf{y}}$ according to multiple feature functions as

$$\begin{aligned}\hat{\mathbf{y}} &= \arg\max_{\mathbf{y}} p\left(\mathbf{y}|\mathbf{x}\right) \\ &= \arg\max_{\mathbf{y}} \left\{ \frac{\exp(\sum_{m=1}^{M} \lambda_m h_m\left(\mathbf{y}, \mathbf{x}\right))}{\sum_{\mathbf{y}'} \exp(\sum_{m=1}^{M} \lambda_m h_m\left(\mathbf{y}', \mathbf{x}\right))} \right\} \\ &= \arg\max_{\mathbf{y}} \left\{ \exp\left(\sum_{m=1}^{M} \lambda_m h_m\left(\mathbf{y}, \mathbf{x}\right) \right) \right\} \\ &= \arg\max_{\mathbf{y}} \left\{ \sum_{m=1}^{M} \lambda_m h_m\left(\mathbf{y}, \mathbf{x}\right) \right\}\end{aligned} \tag{8}$$

where $h_m\left(\mathbf{y}, \mathbf{x}\right)$ are the feature functions and λ_m are the corresponding weights. The following SMT feature functions are typically used: inverse phrase translation probability, inverse lexical weighting, direct phrase translation probability, direct lexical weighting, the language model, the distance-based reordering model, the word penalty. Neural network language models [34]–[36] can also be used as an additional features, which generally leads to a significant improvement.

The goal of model tuning is to find the optimal weights for the log-linear model. The weights can be tuned by the Minimum Error Rate Training (MERT) algorithm [37] or Margin Infused Relaxed Algorithm (MIRA) [38]. During translation, the SMT decoder selects a target segment (word or phrase) translation for an uncovered source segment, to expand the partial translation $\mathbf{y_{<t}} = y_1, y_2, \ldots, y_{t-1}$. The translation process finishes after all the source segments are translated.

III. The Proposed Approach

This section describes our framework that incorporates the SMT word knowledge into the NMT word generation process, as well as two proposed model variants to implement the framework. The word prediction procedure is the most different part between the conventional attention-based NMT and the proposed model. The proposed model predicts target words based

not only on conventional NMT word prediction but also on SMT word knowledge. In the following subsections, we will first describe the generation process of SMT word recommendations on individual decoding step. Then we will describe the integration of SMT word recommendations into the NMT encoder-decoder architecture. Finally, we will introduce a simple yet effective method to overcome the UNK problem using the SMT word recomendations.

A. Generating SMT Word Recommendations

In order to be compatible with NMT word-by-word generation style, in this work we adopt a word-based simplification of the general SMT phrase-based model to make SMT word recommendations.

At each decoding step, the SMT model uses the NMT decoding information to make word recommendations. Given previously generated words $\mathbf{y}_{<\mathbf{t}} = y_1, y_2, \ldots, y_{t-1}$ from NMT, SMT expands the partial translation $\mathbf{y}_{<\mathbf{t}}$ and generates the next word recommendations according to their scores, which is defined as follows

$$SMT_{score}(y_t|\mathbf{y}_{<\mathbf{t}}, \mathbf{x}) = \sum_{m=1}^{M} \lambda_m h_m(y_t, x_t) \tag{9}$$

where y_t is a generated word and x_t is its corresponding source span. $h_m(y_t, x_t)$ are the feature functions and λ_m are the corresponding weights. Ideally the SMT model takes the words with the high scores as word recommendations to guide NMT word prediction. However, there are two challenges for generating SMT word recommendations due to the different translation mechanisms between the two sorts of models.

Coverage Information. SMT model cannot record which source words are covered, due to NMT generates target sentence without explicit source coverage information.[1] The NMT translation process is complete when the end of sentence symbol "EOS" is generated. In contrast, the SMT decoder explicitly utilizes a coverage vector to indicate whether a source word is translated or not. The SMT translation process is finished when all source words are translated.

The coverage information is missing since the SMT decoder cannot obtain explicit source coverage information from NMT decoding information, which results in generating inappropriate word recommendations. To address the issue, we introduce an SMT coverage vector $\mathbf{cv} = [cv_1, cv_2, \ldots, cv_{T_x}]$ for the SMT decoder. Specifically, $cv_i = 1$ means that the source word x_i has been translated by the proposed model and the SMT decoder will not generate word recommendations from the source word which is labeled $cv_i = 1$.

We initialize the SMT coverage vector to a zero vector. At each decoding step, if the generated word is in current SMT word recommendations, the corresponding element in SMT coverage vector will be updated. Specifically, we first use explicit SMT word alignments which are generated with an unsupervised aligner to locate the source word which aligns to the used word recommendation, then we update the vector by setting the corresponding element to one.

Alignment Information. Word order differences exist between different types of languages, therefore the reordering model plays a crucial role in existing SMT approaches. Typically, the SMT model uses the word alignment information to calculate the reordering score. For example, the SMT model calculates a distance-based reordering score as follows [17]:

$$d(y_t) = -\left|sp_{y_t} - sp_{y_{t-1}} - 1\right| \tag{10}$$

where sp_y denotes the position of source word which is aligned to target word y.

As described in Section II, NMT uses attention weights to indicate how well a target word and a source word align. The attention weights can be treated as word alignments in SMT approaches [40], [41]. However, it differs from conventional SMT alignments which align the target word to specific source words. NMT uses normalized attention probabilities over the entire source sentence to present the alignment information, instead of "attending" to a specific source word. Therefore the conventional reordering model cannot directly work with normalized attention probabilities produced by the attention model in Equation (6).

To overcome this problem, we modify Equation (10) to make use of normalized attention probabilities over the entire sentence to calculate the SMT reordering score. Similar to the estimation of bi-directional word translation probabilities in [29], the modified reordering score is calculated as follows:

$$d(y_t) = -\sum_{j=1}^{T_x} \alpha_{t-1,j} \left|sp_{y_t} - j - 1\right| \tag{11}$$

where $\alpha_{t-1,j}$ are alignment weights generated by the NMT attention model. The lexical reordering model can work in a similar way.

Considering the second disadvantage of NMT as mentioned in Section I, we filter out stop words from SMT word recommendations, to keep content word recommendations which convey the meaning of the untranslated source word.

B. Integrating SMT Word Recommendations Into NMT

We adopt a neural network-based classifier to digest the SMT word recommendations. The classifier uses the SMT word recommendations and current decoding state of the NMT model to generate probability estimates for SMT word recommendations. The generated probabilities indicate how well a SMT word recommendation and the current decoding state match. To reduce the burden of the classifier, we propose to use heuristic methods to filter the low-quality SMT word recommendations: 1) local limitation. For each source word, we calculate the translation scores[2] of these word recommendations and choose the highest N_{local} as recommendation candidates. 2) global limitation. We

[1] [24] propose to introduce a linguistic coverage model to NMT model. It is complementary to our proposed model in dealing with coverage problem. [39] propose to use the phrase-based SMT model to rerank the n-best NMT outputs. Specifically, they propose a soft forced decoding algorithm that can match a NMT output to an SMT decoding path. Forced decoding algorithm can be introduced to compute a score for the generated SMT word recommendations.

[2] Bidirectional translation probabilities and bidirectional lexical translation probabilities.

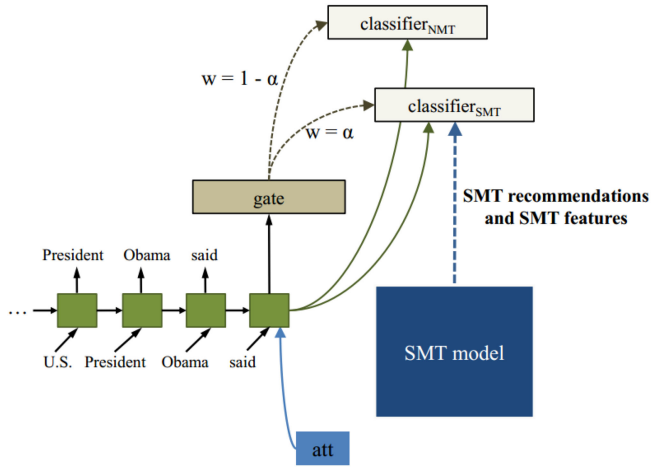

Fig. 1. Architecture of the decoder using the gating mechanism in NMT with SMT word recommendations. The blue box with characters "att" is the context vector outputted by the encoder of NMT with attention mechanism.

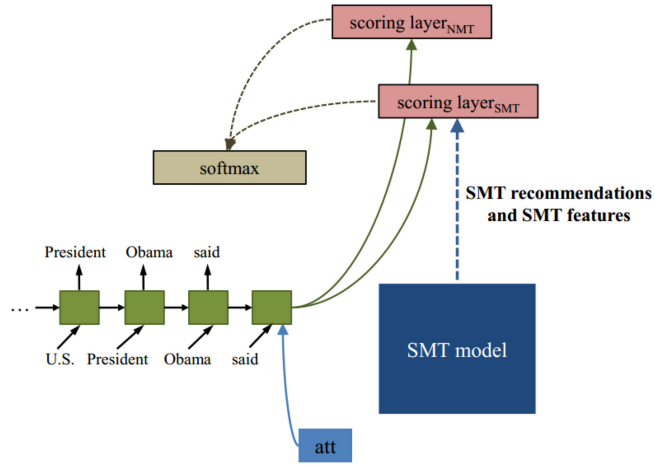

Fig. 2. Architecture of the decoder using the direct competition mechanism in NMT with SMT word recommendations. The blue box with characters "att" is the context vector outputted by the encoder of NMT with attention mechanism.

choose the highest N_{global} word recommendations according to SMT scores, which are calculated as a weighted sum of SMT features according to Equation (9) and Equation (11), among all local recommendation candidates as final word recommendations.

In this work, we use two model variants, *gating mechanism* and *direct competition mechanism* to estimate and combine the probabilities of SMT word recommendations and NMT word prediction. In the following paragraphs, we detail the estimation and combination of the word probabilities.

Gating Mechanism. The architecture of the decoder using gating mechanism is shown in Figure 1. The decoder with integration of the SMT word knowledge generates word as follows:

1) The conventional NMT classifier (i.e., "classifier$_{\mathtt{NMT}}$") uses current hidden state, the previous generated target word and the current context vector to make word prediction:

$$p_{nmt}(y_t|\mathbf{y}_{<\mathbf{t}},\mathbf{x}) = softmax\left(f(s_t, y_{t-1}, c_t)\right) \quad (12)$$

2) The neural network-based classifier (i.e., "classifier$_{\mathtt{SMT}}$") estimates probabilities on the SMT word recommendations, which are generated by the auxiliary SMT model.
3) The proposed model uses the gating mechanism (i.e., "gate" in brown box in Fig. 1) to combine the word probabilities in the above two steps, to calculate the posterior probability.

At decoding step t, the neural network-based classifier (i.e., "classifier$_{\mathtt{SMT}}$") estimates probabilities on the SMT word recommendation vocabulary V_t^{smt} as follows:

$$p_{smt}(y_t|\mathbf{y}_{<\mathbf{t}},\mathbf{x}) = softmax\left(score_{smt}(y_t|\mathbf{y}_{<\mathbf{t}},\mathbf{x})\right) \quad (13)$$

where y_t is a SMT word recommendation and $score_{smt}(y_t|\mathbf{y}_{<\mathbf{t}},\mathbf{x})$ is the scoring function defined as follows:

$$score_{smt}(y_t|\mathbf{y}_{<\mathbf{t}},\mathbf{x}) = g_{smt}(s_t, y_{t-1}, y_t, c_t) \quad (14)$$

where $g_{smt}(\cdot)$ is an activation function which is either an identity or a non-linear function.

At last, we introduce the gating mechanism to calculate the posterior prediction probability on regular vocabulary V^{nmt}. Specifically, we perform a linear interpolation between two probabilities of the NMT generations and SMT word recommendations, to update the prediction probability:

$$p(y_t|\mathbf{y}_{<\mathbf{t}},\mathbf{x}) = (1-\alpha_t)p_{nmt}(y_t|\mathbf{y}_{<\mathbf{t}},\mathbf{x}) + \alpha_t p_{smt}(y_t|\mathbf{y}_{<\mathbf{t}},\mathbf{x}) \quad (15)$$

Note that $p_{smt}(y_t|\mathbf{y}_{<\mathbf{t}},\mathbf{x}) = 0$ for $y_t \notin V_t^{smt}$, as the SMT word recommendation vocabulary at decodint step t V_t^{smt} is N_{global}, which is smaller than the NMT regular vocabulary V^{nmt}. And the gate α_t is calculated as follows:

$$\alpha_t = g_{gate}(s_t, y_{t-1}, c_t) \quad (16)$$

where $g_{gate}(\cdot)$ is the $sigmoid$ function.

Direct Competition Mechanism. The architecture of the decoder using the direct competition mechanism is shown in Figure 2. The decoder with integration of the SMT word knowledge generates word as follows:

1) The NMT scoring layer (i.e., "scoring layer$_{\mathtt{NMT}}$") scores the word predictions on the regular vocabulary V^{nmt}.
2) The SMT scoring layer (i.e., "scoring layer$_{\mathtt{SMT}}$") scores the SMT word recommendations, which are generated by the auxiliary SMT model.
3) The proposed model estimates the probabilities of the SMT word recommendations and the NMT word generations through direct competition over their scores estimated in the above two steps (i.e., "softmax" in brown box in Fig. 2).
4) The proposed model uses the direct competition mechanism to combine the word probabilities, to calculate the posterior probability.

At decoding step t, the SMT scoring layer generates probabilities on the SMT word recommendations as follows:

$$score_{smt}(y_t|\mathbf{y}_{<\mathbf{t}},\mathbf{x}) = g_{smt}(s_t, y_{t-1}, y_t, c_t) \quad (17)$$

where $g_{smt}(\cdot)$ is a non-linear activation function.

The proposed model updates the word prediction probability by the "mixture" of probabilities as follows

$$p(y_t|\mathbf{y_{<t}}, \mathbf{x}) = p_{nmt}(y_t|\mathbf{y_{<t}}, \mathbf{x}) + p_{smt}(y_t|\mathbf{y_{<t}}, \mathbf{x}) \quad (18)$$

where

$$p_{nmt}(y_t|\mathbf{y_{<t}}, \mathbf{x}) = \begin{cases} \frac{1}{Z} e^{score_{nmt}(y_t|\mathbf{y_{<t}}, \mathbf{x})} & y_t \in V^{nmt} \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

$$p_{smt}(y_t|\mathbf{y_{<t}}, \mathbf{x}) = \begin{cases} \frac{1}{Z} e^{score_{smt}(y_t|\mathbf{y_{<t}}, \mathbf{x})} & y_t \in V_t^{smt} \\ 0 & \text{otherwise} \end{cases} \quad (20)$$

where the normalization term Z shared by the SMT and NMT models is computed as

$$Z = \sum_{y_t \in V^{nmt}} e^{score_{nmt}(y_t|\mathbf{y_{<t}}, \mathbf{x})} + \sum_{y_t \in V_t^{smt}} e^{score_{smt}(y_t|\mathbf{y_{<t}}, \mathbf{x})} \quad (21)$$

Due to the shared normalization term, the SMT word recommendations are in direct competition with NMT word generations through a softmax function.

C. Handling UNK With SMT Word Recommendations

The SMT model makes word recommendations without UNK symbols. However, the neural network-based classifier adopts the word recommendations still map the rate words into UNK symbol. To overcome the UNK issue, we propose to select a proper SMT word recommendation to replace the UNK symbol. Specifically, when UNK symbol is generated, the proposed model switches to copy the word recommendation with highest SMT score to replace the UNK symbol.

IV. Model Training

We first separately train the SMT model on a parallel corpus using the conventional SMT approach [17], then we train the proposed model to minimize the negative log-likelihood:

$$C(\theta) = -\frac{1}{N_{train}} \sum_{n=1}^{N_{train}} \sum_{t=1}^{T_y} \log p(y_t^n|\mathbf{y^n_{<t}}, \mathbf{x^n}) \quad (22)$$

given the training data with N_{train} bilingual sentences $\{(\mathbf{x^1}, \mathbf{y^1}), (\mathbf{x^2}, \mathbf{y^2}), \ldots, (\mathbf{x^{N_{train}}}, \mathbf{y^{N_{train}}})\}$ [42]. The conditional probability $p(y_t^n|\mathbf{y^n_{<t}}, \mathbf{x^n})$ is computed using Equation (15) or Equation (18).

As shown in Equation (11), the quality of attention weights plays an important role in generating appropriate SMT word recommendations. The low quality attention weights may result in generating inappropriate recommendations, which negatively affect the proposed model training. To overcome the issue, we adopt a simple pre-training strategy for training the proposed model. We first train a regular attention-based NMT model. The shared parameters between the proposed model and the regular attention-based NMT model are initialized by the fine-trained baseline model. The new parameters of the proposed model are randomly initialized. Finally, all of the parameters of the proposed model are trained to minimize the cost function.

V. Experiments

In this section, we examine the performance of the proposed approach on Chinese-to-English and English-to-German machine translation tasks and do some case studies for our approach.

A. Setup

For Chinese-to-English, the bilingual training data consists of about 1.25 million sentence pairs[3] with 27.9 million Chinese words and 34.5 million English words respectively. We use NIST2006 (NIST06) dataset for development, and NIST2002 (NIST02), NIST2003 (NIST03), NIST2004 (NIST04), NIST2005 (NIST05) and NIST2008 (NIST08) datasets for evaluation. We report experiment results with case insensitive 4-gram BLEU score[4] for evaluation.

For English-to-German, to compare with the previous work, we use the same subset of the WMT2014[5] training set, containing 4.0 million sentence pairs with 91.1 million English words and 86.6 million German words. The concatenation of news-test 2012 and news-test 2013 is used for development and news-test 2014 is used for evaluation. We report experiment results with case sensitive BLEU score[6] for evaluation.

We compare our proposed model with state-of-the-art translation systems:

* Moses: a state-of-the-art phrase-based SMT system.
* RNNSearch: an attention-based NMT system. We use the open source system GroundHog[7] as the NMT baseline.

For Moses, we use the default setting on the entire training data to train the model, and use KenLM[8] [43] on the target portion of the bilingual training data to train a 4-gram language model. As for English-to-German, we also expand the language model training data by using the available German monolingual training data (Europarl v7) from the shared task,[9] which contains 2.2M sentence with 47.2M German words. We run Giza++[10] on the training corpus in both source-to-target and target-to-source directions and apply the "grow-diag-final" refinement rule [17] to obtain word alignments. We adopt MERT [37] to tune feature weights of the Moses decoder.

For RNNSearch, we use the bilingual data to train the NMT baseline model. We set the source and target vocabularies to 30,000 most frequent words for Chinese-to-English and 50,000

[3]The corpus includes LDC2002E18, LDC2003E07, LDC2003E14, Hansards portion of LDC2004T07, LDC2004T08 and LDC2005T06.

[4]ftp://jaguar.ncsl.nist.gov/mt/resources/mteval-v11b.pl

[5]http://www.statmt.org/wmt14/translation-task.html

[6]https://github.com/moses-smt/mosesdecoder/blob/master/scripts/generic/multi-bleu.perl

[7]https://github.com/lisa-groundhog/GroundHog

[8]https://kheafield.com/code/kenlm/

[9]http://www.statmt.org/wmt14/translation-task.html

[10]http://www.statmt.org/moses/giza/GIZA++.html

most frequent words for English-to-German, covering approximately 97.7% and 99.3% of tokens in Chinese-to-English and 97.6% and 93.7% of tokens in English-to-German, respectively. We map out-of-vocabulary words to a special token UNK. We train the model with the sentence pairs whose source and target lengths are shorter than 50 in training data and keep the test data at the original length. The word embeddings of the encoder and decoder are set to 620 and the hidden layers of the encoder and the decoder have 1000 hidden units. All the other settings are the same as in [9].

We implemented NMT models using Python in the Theano framework [44]. We also use our in-house implementation of RNNSEARCH as an additional NMT baseline system. We adopt dropout [45] technique for our in-house implementation of RNNSEARCH to enhance the system performance. Dropout is applied only on the output layer with a dropout rate of 0.5. During training, we use Adadelta [46] with the decay rates $\rho = 0.95$ and $\epsilon = 10^{-6}$ to optimize the model, and each batch contains 80 sentences. During testing we use a beam size of 10 for beam search.

For SMT word recommendation, we re-implement an SMT decoder which only adopts bidirectional translation features, bidirectional lexical translation features, language model feature and distance-based reordering feature to compute the SMT score according to Equation (9). The distance-based reordering feature is calculated according to Equation (11). Lexical reordering features are not used in order to speed up the SMT word recommendation process. As for word-level recommendation, word penalty feature and phrase penalty feature are also abandoned as they cannot provide useful information for calculating recommendation scores. The feature weights are obtained from Moses.[11] We add punctuations into the stop word list and remove numerals from the list. Finally the target language of English and German stop lists contain 572 symbols and 849 symbols respectively. To ensure the quality of SMT word recommendations, we set N_{local} to 5 and N_{global} to 25. We adopt a forward neural network with two hidden layers for the SMT classifier (Equation (14)), the SMT scoring layer (Equation (17)) and gating function (Equation (16)). The numbers of units in the hidden layers are 200 and 50 respectively.

B. Results on Chinese-to-English Translation

From Table II, we observe that attention-based NMT baseline system RNNSEARCH outperforms Groundhog and the SMT baseline Moses by 1.40 BLEU points and 0.66 BLEU points. Experimental results show that two variants of the proposed model significantly outperform the NMT baseline and the SMT baseline systems.

Specifically, RNNSearch$_{+\texttt{SMTrec/gate}}$ with integration of the SMT word knowledge in gating mechanism but keeps target UNK words obtains a gain of up to 1.21 BLEU points over the baseline system RNNSEARCH. Surprisingly, RNNSearch$_{+\texttt{SMTrec/gate, UNK Replace}}$ which further uses the SMT word recommendation to replace UNK during translation, shows further improvement over RNNSearch$_{+\texttt{SMTrec/gate}}$ by 1.35 BLEU points. It finally outperforms RNNSEARCH and Moses by 2.56 BLEU points and 3.22 BLEU points respectively. RNNSearch$_{+\texttt{SMTrec/softmax}}$ with integration of the SMT word knowledge in direct competition mechanism but keeps target UNK words achieves 1.10 BLEU points improvement over the RNNSEARCH on average, and RNNSearch$_{+\texttt{SMTrec/softmax, UNK Replace}}$ achieves absolute improvements of 2.49 and 3.15 BLEU points over RNNSEARCH and Moses respectively.

Integrating SMT word knowledge into the NMT decoder decreases the decoding speed slightly. The decoding speed of the baseline system RNNSEARCH and our proposed model RNNSearch$_{+\texttt{SMTrec/gate}}$ are 16.7 and 13.2 words per second respectively.[12]

We also conduct additional experiments to validate the effectiveness of the key components of our proposed model. ARTHUR is the method which incorporates discrete translation lexicons into NMT model by making use of attention alignment probability [25]. RNNSearch$_{+\texttt{SMTrec/gate withoutreordering}}$ is the proposed model which uses SMT features except the reordering feature. RNNSearch$_{+\texttt{SMTrec/gate, attention Replace}}$ uses NMT attention alignments to implement the UNK replacement. It uses the largest α in Equation (6) to infer the source word position for target UNK symbol [23] and replaces the UNK symbol with the translation of the source word in a dictionary, where the dictionary is constructed from the SMT word alignments [28]. We apply this method on the output generated by RNNSearch$_{+\texttt{SMTrec/gate}}$.

From Table III, we can observe that:

1) ARTHUR outperforms RNNSEARCH by 0.85 BLEU points, while our proposed model RNNSearch$_{+\texttt{SMTrec/gate}}$ outperforms RNNSEARCH by 1.21 BLEU points. This indicates that SMT features are useful to generate word knowledge which improves NMT translation performance.
2) RNNSearch$_{+\texttt{SMTrec/gate withoutreordering}}$ improves the baseline RNNSEARCH with 0.80 BLEU points. Through the comparison of RNNSearch$_{+\texttt{SMTrec/gate}}$ performance that obtains an improvement of 1.21 BLEU socres, it is clear that reordering feature which plays a critical role in SMT is very important for generating proper SMT word recommendations.
3) RNNSearch$_{+\texttt{SMTrec/gate, attention Replace}}$ obtains an improvemnt over RNNSearch$_{+\texttt{SMTrec/gate}}$ with 0.87 BLEU point, while RNNSearch$_{+\texttt{SMTrec/gate, UNK Replace}}$ obtains improvements with 1.35 BLEU points. This demonstrates that handling UNK with SMT word recommendations is effective.

At last, we use the subword-based system as the baseline system to evaluate our proposed model. We use Byte Pair Encoding (BPE) [47] preprocessing scripts[13] with 30,000 operations to segment the Chinese and English sentences into subword units respectively. The baseline system RNNSEARCH and the SMT

[11]Compared with other weights, the weight of distance-based reordering feature is too small, therefore we manually increase it by 10 times.

[12]The experiments are conducted on a single NVIDIA Tesla K80 GPU.

[13]https://github.com/rsennrich/subword-nmt

TABLE II
CHINESE-TO-ENGLISH TRANSLATION RESULTS. "$*$" AND "$**$" INDICATE STATISTICALLY SIGNIFICANTLY BETTER THAN RNNSEARCH WITH $p < 0.05$ AND $p < 0.01$ RESPECTIVELY

SYSTEM	NIST06	NIST02	NIST03	NIST04	NIST05	NIST08	Avg
MOSES	32.43	34.08	34.16	34.74	31.99	23.69	31.73
GROUNDHOG	30.23	34.79	32.21	34.02	30.56	23.37	30.99
RNNSEARCH	33.66	34.82	32.92	36.33	32.91	24.98	32.39
+SMTrec/softmax	34.25	35.83	33.86**	37.98**	34.16**	25.60	33.49
+SMTrec/softmax, UNK Replace	34.53**	38.09**	36.15**	38.53**	35.84**	25.80**	34.88
+SMTrec/gate	34.40*	36.11*	34.15**	37.18**	33.75**	26.80**	33.60
+SMTrec/gate, UNK Replace	**34.76****	**37.84****	**35.98****	**38.45****	**35.38****	**27.12****	**34.95**

TABLE III
CHINESE-TO-ENGLISH TRANSLATION RESULTS

SYSTEM	NIST06	NIST02	NIST03	NIST04	NIST05	NIST08	Avg
GROUNDHOG	30.23	34.79	32.21	34.02	30.56	23.37	30.99
ARTHUR	34.28	35.97	33.86	36.98	33.68	25.72	33.24
RNNSEARCH	33.66	34.82	32.92	36.33	32.91	24.98	32.39
+SMTrec/gate	34.40	36.11	34.15	37.18	33.75	26.80	33.60
+SMTrec/gate without reordering	34.04	35.65	33.97	36.82	33.91	25.59	33.19
+SMTrec/gate, UNK Replace	**34.76**	**37.84**	**35.98**	**38.45**	**35.38**	**27.12**	**34.95**
+SMTrec/gate, attention Replace	34.66	37.04	35.25	38.10	34.93	27.03	34.47
RNNSEARCH+subword	35.09	36.56	34.61	37.98	34.50	26.66	34.06
+SMTrec/gate	35.74	37.63	35.74	39.02	35.49	27.48	35.07

TABLE IV
CHINESE-TO-ENGLISH TRANSLATION RESULTS. "$*$" AND "$**$" INDICATE STATISTICALLY SIGNIFICANTLY BETTER THAN RNNSEARCH-COVERAGE WITH $p < 0.05$ AND $p < 0.01$ RESPECTIVELY

SYSTEM	NIST06	NIST02	NIST03	NIST04	NIST05	NIST08	Avg
RNNSEARCH	33.66	34.82	32.92	36.33	32.91	24.98	32.39
RNNSEARCH-COVERAGE	34.89	36.28	33.90	37.42	34.10	26.23	33.59
+SMTrec/softmax	35.10**	36.91*	35.12**	38.14**	34.83*	26.48**	34.30
+SMTrec/softmax, UNK Replace	35.38**	39.23**	37.62**	39.04**	36.13**	26.62**	35.73
+SMTrec/gate	35.34**	37.36**	35.20**	38.57**	35.34**	26.54**	34.60
+SMTrec/gate, UNK Replace	35.57**	39.25**	37.32**	39.26**	36.54**	26.66**	35.81

model are trained on the subword-based bilingual data. At each decoding step, the SMT model generates word/subword knowledge to guide NMT decoding. The experiment results are shown in Table III. We find that the proposed model also outperforms the subword-based baseline system by 1.01 BLEU points.

Coverage-based NMT [24] is a variant of attention-based NMT model, which introduces a coverage mechanism into the conventional attention-based NMT model. We further adopt neural network based coverage model[14] as our strong baseline system, which we denote as RNNSEARCH-COVERAGE, and integrate the proposed models into RNNSEARCH-COVERAGE. We also adopt pre-training strategy described in Section IV to train the model. More specifically, we train the coverage-based NMT model and use its parameters to initialize the corresponding part of our proposed model. Experiment results are shown in Table IV. The experimental results demonstrate that RNNSEARCH-COVERAGE outperforms the baseline NMT system RNNSEARCH and the combination of coverage-based model and the proposed model yields further improvements.

[14]Please refer to [24] for more details.

TABLE V
NUMBERS OF CONTENT WORDS IN NIST08 GENERATED BY RNNSEARCH AND RNNSEARCH$_{+\texttt{SMTrec/gate}}$

	ALL	REMOVE
Reference	20481	19489
RNNSEARCH	13230	11007
+SMTrec/gate	12665	11172

C. Analysis on Generated Words

We also check the number of content words in NIST08 test set generated by the baseline system and RNNSearch$_{+\texttt{SMTrec/gate}}$, as shown in Table V. The numbers of content words produced by the baseline system and RNNSearch$_{+\texttt{SMTrec/gate}}$ are 13230 and 12665 respectively. Then we remove duplicates in the generated content words, we find 11007 and 11172 content words in the baseline system and RNNSearch$_{+\texttt{SMTrec/gate}}$ respectively. These numbers indicates that two systems generate over-translation. Concerning the number of duplicates in the references ($20481 - 19489 = 992$), we find that RNNSearch$_{+\texttt{SMTrec/gate}}$($12665 - 11172 = 1493$) makes less mistakes on "over-translation" problem than the baseline system ($13230 - 11007 = 2223$).

TABLE VI
CHINESE-TO-ENGLISH TRANSLATION RESULTS ON LARGE-SCALE CORPUS. RNNSEARCH IS OUR NMT SYSTEM, AND RNNSEARCH-COVERAGE IS A STATE-OF-THE-ART VARIANT OF ATTENTION-BASED NMT MODEL [24]. "$*$" AND "$**$" INDICATE STATISTICALLY SIGNIFICANTLY BETTER THAN RNNSEARCH-COVERAGE WITH $p < 0.05$ AND $p < 0.01$ RESPECTIVELY

SYSTEM	NIST06	NIST02	NIST03	NIST04	NIST05	NIST08	Avg
RNNSEARCH	37.17	38.04	36.67	39.82	36.77	28.82	36.02
RNNSEARCH-COVERAGE	37.66	38.99	37.78	41.52	38.49	28.87	37.13
+SMTrec/gate	37.91**	40.43**	38.61**	42.54**	39.75**	29.31**	38.13
+SMTrec/gate, UNK Replace	**38.11****	**42.61****	**41.22****	**42.78****	**40.39****	**29.45****	39.29

TABLE VII
COMPARISON WITH PREVIOUS WORK ON WMT ENGLISH-TO-GERMAN TRANSLATION. THE BLEU SCORES ARE CASE SENSITIVE

SYSTEM	BLEU
Existing end-to-end NMT baseline [23]	16.46
WMT'14 winner system Phrase-based SMT [48]	20.67
Our work	
Moses	18.06
RNNSEARCH	16.98
+SMTrec/gate	18.22
+SMTrec/gate, UNK Replace	19.21

D. Results on Large-Scale Chinese-to-English Translation

To further validate the proposed framework, we also carry out experiments on a large-scale bilingual training data from LDC. The training data contains 5.3 million sentence pairs with approximately 102.2 million Chinese words and 116.6 million English words. The vocabularies are enlarged to 40,000 words both in Chinese and English, which covers approximately 98.6% and 99.3% of Chinese and English words in training data respectively. All the other settings are the same as the NMT experiments in Subsection A. Experiment results are detail in Table VI. The results in Table VI further verify the effectiveness of our proposed models.

E. Results on English-to-German Translation

Table VII shows the experimental results on English-to-German translation. From the table, we can find that our proposed model also outperforms the NMT and SMT baseline systems on English-to-German translation in terms of BLEU score. The experimental results demonstrate that our proposed model is language independent.

F. Case Study

In this section, we show several translation examples in Chinese-to-English translation to illustrate why the proposed model can improve translation quality.

As our method of handling UNK achieves significant improvement, we first show two translation examples in Table VIII to illustrate the translation improvement. During translation, the NMT model always encounters source word which is out of its vocabulary and generates a target UNK word. There are two types of source words which single NMT model cannot translate. As shown in Table VIII, one type is the word that is not in NMT regular vocabulary (UNK), the other type is the word that has not appeared in bilingual corpus (OOV). Our method handle the former one by selecting a proper SMT word recommendation. We observe that most of UNK words are named entities and numbers. A right translation replacement may earn a big bonus for BLEU score as it simultaneously improves 1-gram/2-gram/3-gram/4-gram BLEU scores. However, for the OOV word, e.g., Chinese words "Insert Graphics" and "Insert Graphics" in Table IX, our method fails to translate it as the OOV word is also a difficult problem for the conventional SMT approaches.

Then we show a translation example in Table IX to illustrate the translation improvement made by the SMT word recommendations. In this example, RNNSEARCH omits to translate the source Chinese word "Insert Graphics", while with the help of the SMT word recommendations (the target word "sir" in Moses output indicates that "sir" is indeed in the SMT word recommendations), the proposed model successfully retrieves the source word and translates it into target word "sir'.

VI. RELATED WORK

Our work is related to three main areas of research: combination of SMT and NMT, coverage problem in NMT and UNK word problem in NMT.

Combination of SMT and NMT: [30] proposed to utilize structured search spaces produced by a hierarchical phrase-based SMT model (Hiero) to guide NMT decoding by expanding the NMT search space with Hiero translation hypotheses. [29] treated the NMT model as an additional feature and associate it with effective SMT features to improve translation performance. [49] enhanced NMT system with SMT outputs. They used SMT model to pre-translate the source sentence into the target language. Then used a NMT model to generate the final hypothesis with the pre-translation as an addition input. [50] combined NMT and SMT outputs to form a neural system combination framework. [25] proposed to incorporate discrete translation lexicons into the NMT model. They first calculated the lexical predictive probability over the next word, and then integrated the probability with the NMT model probability to predict the next word. [51] proposed to integrate SMT phrase memory translation into an NMT model. The SMT model writes relevant target phrases into a phrase memory and the NMT model reads the phrases and selects the proper phrase to generate. [52] proposed a hybrid search for NMT. To introduce phrase translation hypotheses from SMT model, the hybrid search algorithm allows hypothesizing phrases in addition to normal word hypotheses. A log-linear model with an NMT model scores and SMT scores is utilized to score theses hypotheses. [53] presented minimum

TABLE VIII
TRANSLATION EXAMPLES FOR HANDLING UNK WITH SMT WORD RECOMMENDATIONS. UNK MEANS THE WORD IS NOT IN NMT REGULAR VOCABULARY. OOV MEANS THE WORD HAS NOT APPEARED IN BILINGUAL CORPUS

source	拉斯穆森/UNK 说 : 我们 同样 认为 通过 对话 宽容 来 解决 事端 , 而 不 是 暴力 。
reference	rasmussen said, we also think that we should resolve this issue through dialogue and tolerance, not through violence .
Moses	rasmussen said : we also believe resolved through dialogue tolerance , instead of violent incidents .
+SMT rec	UNK said , " we also believe that disputes can be solved through dialogue and tolerance , not violence . "
+SMT rec, UNK Replace	rasmussen said , " we also believe that disputes can be solved through dialogue and tolerance , not violence . "
source	诺桑比亚/OOV 警局/UNK 也 表示 , 没有 证据 显示 这 两 起 窃案 和 史蒂文生/OOV 爵士 的 工作 有关 。
reference	northumbria police also said there was no evidence that the two burglaries were linked to lord stevens' work .
Moses	诺桑比亚 police also indicated that no evidence that these two burglary and sir 史蒂文生 work .
+SMTrec/gate	UNK UNK also said that there was no evidence to show the work of the two burglary and UNK .
+SMTrec/gate, UNK Replace	expressed police also said that there was no evidence to show the work of the two burglary and relevant .

TABLE IX
TRANSLATION EXAMPLES FOR SHOWING IMPROVEMENT MADE BY SMT WORD RECOMMENDATIONS. UNK MEANS THE WORD IS NOT IN NMT REGULAR VOCABULARY. OOV MEANS THE WORD HAS NOT APPEARED IN BILINGUAL CORPUS

source	史蒂文生/OOV 爵士 是 英国 诺桑比亚/OOV 警察局 的 局长 , 他 的 办公室 位 在 泰恩河/OOV 新堡/OOV 一带 。
reference	sir stevens is the head of the northumbria police bureau in the UK. his office is located in the newcastle area on the thames river.
Moses	a british sir 史蒂文生 诺桑比亚 police station of his office , located in the vicinity 泰恩河 新堡 .
RNNSearch	UNK is the secretary of the british police station , and his office is located in the UNK area .
+SMTrec/gate	sir UNK is the secretary for the british UNK police , his office located in the UNK area of UNK .

Bayes risk decoding with PBMT n-gram posteriors to combine NMT with SMT.

The significant difference between our work and these studies is that NMT is treated as a black-box in the previous work, while in our work the NMT and SMT models are tightly integrated with the execution of the former being advised by the latter in the training and testing phase.

Coverage problem in NMT: To alleviate the "over-translation" and "under-translation" problems, [24] proposed to incorporate the coverage mechanism for NMT. The coverage mechanism encourages the NMT model 'attend" more about untranslated source words in the target sentence generation. [54] extended the coverage mechanism by directly modeling translated and untranslated source content, which are fed to both the attention model and decoder states. [11] introduced a coverage penalty into the NMT beam search, to rerank NMT hypothesis. [55] proposed to incorporate structural biases from word based alignment models into NMT, to augment the attention model. [56] proposed to add implicit distortion and fertility models to attention model. [57] improved upon the vanilla attention model by explicitly takes the attention history into consideration. They attached a dynamic memory vector for each source word to keep track of history attention maps and used the dynamic memory to improve the attention model. [58] proposed temporal attention model which memorizes the alignments temporally and aggregates the alignments in the memory allowing the attention model to remember its decisions over past timesteps.

UNK word problem in NMT: [28] proposed several methods to point the position of the source word which is aligned to the UNK word in the target sentence. [23] proposed an efficient approximation based on importance sampling to enlarge the target vocabulary size. [59] proposed a 'substitution-translation-restoration method' to replace rare words with semantic similar in-vocabulary words. Besides, instead of modeling word unit, some work focused on smaller unit modeling [47], [60], especially on character modeling [61]–[64].

VII. CONCLUSION

In this paper, we have shown the framework incorporating SMT word knowledge into NMT encoder-decoder architecture could improve the NMT translation performance. Different from previous work which usually uses SMT knowledge in a 'shadow" manner, our proposed framework integrates the SMT knowledge in a 'deep' manner, where the neural network-based models are used to digest the discrete word knowledge. We use two model variants to implement the framework and experiment results on Chinese-to-English and English-to-German translation tasks have demonstrated the effectiveness of the proposed framework. As the proposed models are modification of

sequence-to-sequence model, they are able to be applied to spoken language translation

There are several directions to improve this work. First, the proposed method would result in a model size larger than the sum of a word-based SMT and NMT models combined. We will investigate model reduction strategies to minimize the model size. Second, the proposed model ignores the feature values of SMT word rec ommendations which are very useful in NMT [29]. Therefore we want to attach information to the SMT word recommendations so as to improve scoring accuracy. Furthermore, we also want to introduce more complicated models to make use of SMT phrase recommendations, where NMT may benefit more from multi-word phrase recommendations than word-level ones as phrase-based SMT is superior to word-based SMT [17].

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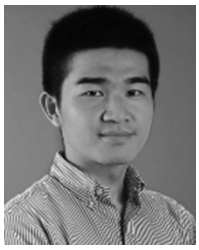

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